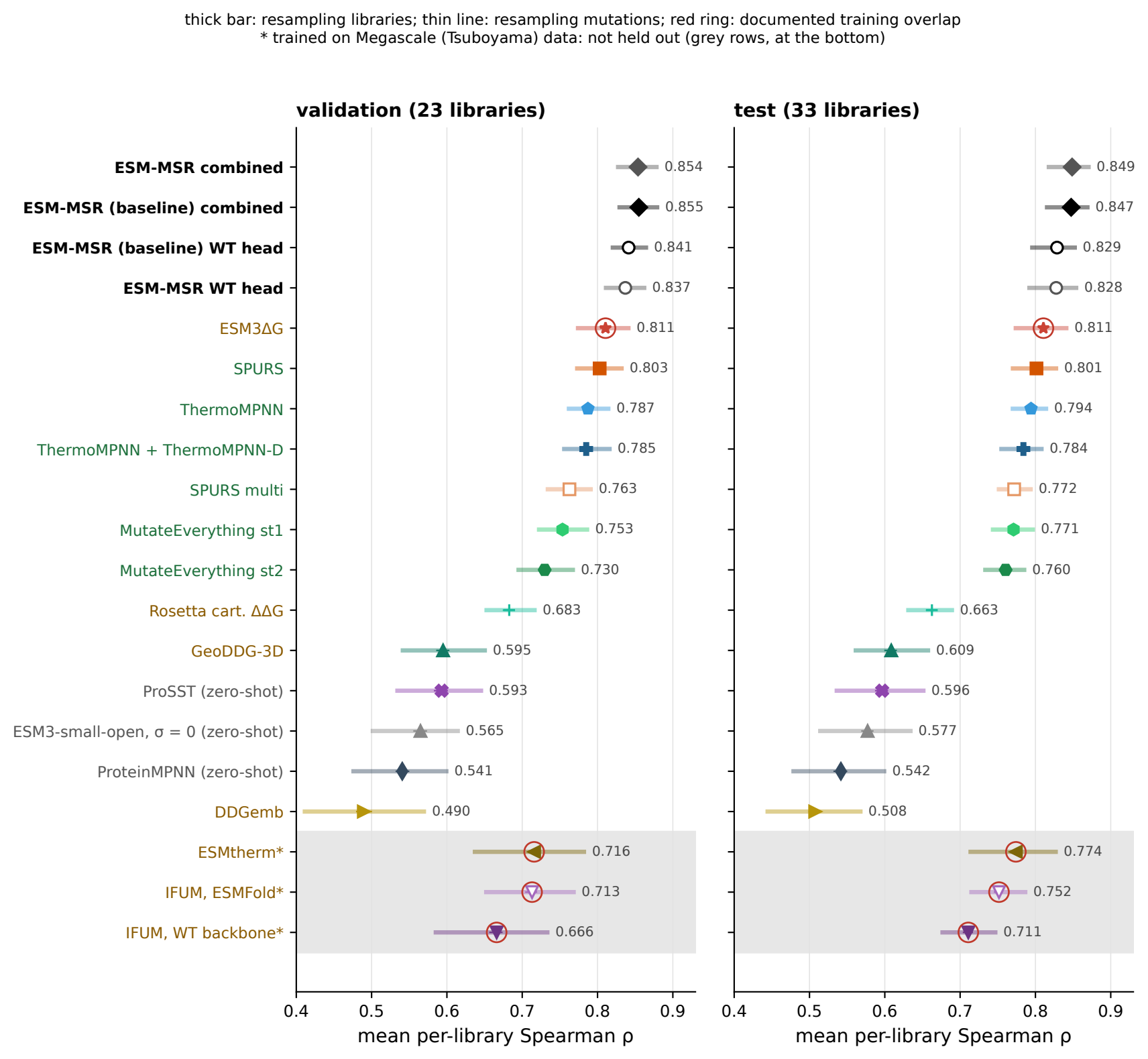
thick bar: resampling libraries; thin line: resampling mutations; red ring: documented training overlap
* trained on Megascale (Tsuboyama) data: not held out (grey rows, at the bottom)
validation (23 libraries)
test (33 libraries)
ESM-MSR combined
ESM-MSR (baseline) combined
ESM-MSR (baseline) WT head
ESM-MSR WT head
ESM3ΔG
SPURS
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
SPURS multi
MutateEverything st1
MutateEverything st2
Rosetta cart. ΔΔG
GeoDDG-3D
ProSST (zero-shot)
ESM3-small-open, σ = 0 (zero-shot)
ProteinMPNN (zero-shot)
DDGemb
ESMtherm*
IFUM, ESMFold*
IFUM, WT backbone*
0.854
0.855
0.841
0.837
0.811
0.803
0.787
0.785
0.763
0.753
0.730
0.683
0.595
0.593
0.565
0.541
0.490
0.716
0.713
0.666
0.849
0.847
0.829
0.828
0.811
0.801
0.794
0.784
0.772
0.771
0.760
0.663
0.609
0.596
0.577
0.542
0.508
0.774
0.752
0.711
0.4
0.5
0.6
0.7
0.8
0.9
mean per-library Spearman ρ